CLASS DISCUSSION PAPER — TEACHER'S COPY (FOR PROJECTION OR HANDOUT)

Mathematics Advanced — Geometric Sequences (GP)

等比数列 • 课堂讨论讲义

Lesson focus: recognising geometric sequences, the common ratio r, the nth term $T_n = ar^{n-1}$, and the sum of the first n terms S_n — with mathematics vocabulary translated into Chinese for dual-language classrooms.

Lesson sequence at a glance

- **1. Vocabulary 词汇** — GP mathematics words, English ↔ Chinese (read them aloud; have students repeat the Chinese and the English).
- **2. What is a GP?** — the constant-ratio test, with three worked examples (doubling, halving, alternating).
- **3. Finding the common ratio** — from consecutive terms and from two non-consecutive terms.
- **4. Finding the nth term** — using $T_n = ar^{n-1}$, including "which term equals ...?".
- **5. Sum of terms** — using S_n and the infinite sum S_∞ (extension).
- **6. Wrap-up** — mixed discussion questions with answers (last page).

How to use it: each section follows rule → worked example → discuss → watch out. The **Discuss** boxes pause the lesson for pair or whole-class talk; the **Watch out** boxes flag the errors seen most often in Exercises 1C and 1D.

Formulas from the NESA Mathematics Advanced Reference Sheet — Sequences and series

nth term (GP): $T_n = ar^{n-1}$ (terms $T_1, T_2, T_3, \ldots$; the first term $a = T_1$ — the reference sheet prints a_n where we write T_n) · Sum of the first n terms: $S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}$ $(r \neq 1)$ · Limiting sum: $S_\infty = \frac{a}{1-r}$ $(|r| < 1)$

The one test that decides everything — 判断等比数列的唯一准则

A sequence is geometric when **the ratio of consecutive terms is constant**: $\frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3} = r$, i.e. $r = \frac{T_2}{T_1} = \frac{T_{n+1}}{T_n}$ (the same identity as in Exercise 1D)

For any three consecutive terms: $T_2^2 = T_1 T_3$ (the square of the middle term equals the product of its neighbours).

1. GP mathematics vocabulary — 等比数列数学词汇

A. Sequences and terms — 数列与项

English	中文	Meaning / example in class
sequence	数列	An ordered list of numbers: 3, 6, 12, 24, ...
term	项	One number of the sequence: the third term here is 12.

English	中文	Meaning / example in class
first term (a)	首项（a）	Where the pattern starts — the first number you write down; $a = T_1$.
geometric sequence / progression (GP)	等比数列	Each term is the previous term multiplied by the same number.
common ratio (r)	公比（r）	The fixed multiplier: $r = T_{n+1} \div T_n$.
arithmetic sequence (contrast)	等差数列	Contrast: add a fixed number each time (a common difference, 公差).
consecutive terms	相邻两项	Terms side by side: T_n and T_{n+1}, e.g. 6 and 12.
term number, n	项数；第 n 项的位置	The position of a term: the 5th term has $n = 5$.

B. Ratio, powers and structure — 比、幂与结构

English	中文	Meaning / example in class
ratio	比，比值	A quotient of two quantities: $12 \div 6 = 2$.
multiply / divide (by r)	乘以 / 除以	Forward: $\times r$ each step. Backward: $\div r$ (divide by the ratio to go back).
constant	常数	Stays the same: in a GP the ratio is constant.
power / index (exponent)	幂；指数	In r^5, the **base** 底数 is r and the index is 5.
index law	指数运算法则	$r^m \times r^n = r^{m+n}$ and $r^m \div r^n = r^{m-n}$ — the engine of every GP calculation.
even power gives two signs	偶次幂给出两个解（±）	$r^4 = 81$ gives $r = \pm 3$ — do not drop the negative.
recurrence relation	递推关系	$T_{n+1} = r\,T_n$ — “multiply by r each time”.
geometric mean	等比中项	b is the geometric mean of a and c when $b^2 = ac$, so $b = \pm\sqrt{(ac)}$.
increasing / decreasing	递增 / 递减	$r > 1$ grows; $0 < r < 1$ shrinks.
alternating	正负交替	A negative ratio flips the sign every term: 5, −15, 45, …
growth / decay	增长 / 衰减	A 3% loss each year means $r = 0.97$ — a decaying GP.
doubling / halving	翻倍 / 减半	The two simplest ratios: $r = 2$ and $r = ½$.

C. Series and applications — 级数与应用

English	中文	Meaning / example in class
sum; **series**	和；级数	Adding the terms: 2 + 6 + 18 is the sum of the first three terms.
sum of the first n terms (S_n)	前 n 项和（S_n）	$S_n = a(1 - r^n) \div (1 - r)$, $r \neq 1$.
finite / infinite	有限 / 无限	A finite GP stops; an infinite GP continues forever.
infinite geometric series (S_∞)	无穷等比级数	$S_\infty = a \div (1 - r)$, but only when $\lvert r\rvert < 1$.
convergent / divergent	收敛 / 发散	The limiting sum exists (converges) only when $\lvert r\rvert < 1$.

English	中文	Meaning / example in class
compound interest	复利	Money growing by a fixed percentage each period is a GP: $A = P(1 + r)^n$.
logarithm	对数	Used when the unknown is in the index: solve $r^n = k$ by $n = \log k \div \log r$.
verify / check	验证 / 检验	Substitute back to test the pattern — always the last step.

中文术语采用简体字（Chinese terms are given in Simplified characters）. **Call-and-response idea:** teacher says the English term, the class answers with the Chinese term (and the meaning), then swap.

2. What is a geometric sequence? — 什么是等比数列?

The constant-ratio test 比值检验法 Divide each term by the term before it. If the answer is always the same number, the sequence is geometric and that number is the **common ratio r**. For three terms: $T_2^2 = T_1T_3$.

Example 1 — a GP that doubles (r = 2)

Show that 3, 6, 12, 24, ... is geometric, and write the next three terms.

$\frac{6}{3} = 2$, $\frac{12}{6} = 2$, $\frac{24}{12} = 2$ — the ratio is constantly 2, so it is a GP.

First term a = 3, common ratio r = 2.

Next three terms: $24 \times 2 = 48$, $48 \times 2 = 96$, $96 \times 2 = 192$.

∴ it is a GP with a = 3, r = 2; the next three terms are 48, 96, 192.

Example 2 — a GP that halves (r = ½)

Show that 80, 40, 20, 10, ... is geometric, and write the next three terms.

$\frac{40}{80} = \frac{1}{2}$, $\frac{20}{40} = \frac{1}{2}$, $\frac{10}{20} = \frac{1}{2}$ — constant, so it is a GP.

Next three terms: $10 \times ½ = 5$, $5 \times ½ = \frac{5}{2}$, $\frac{5}{2} \times ½ = \frac{5}{4}$ (that is, 5, 2.5, 1.25 as decimals — $2.5 = \frac{5}{2}$, $1.25 = \frac{5}{4}$).

∴ it is a GP with a = 80, r = ½; the next three terms are 5, $\frac{5}{2}$, $\frac{5}{4}$.

Example 3 — an alternating GP (r = −3)

Show that 5, −15, 45, −135, ... is geometric, and write the next three terms.

$\frac{-15}{5} = -3$, $\frac{45}{-15} = -3$, $\frac{-135}{45} = -3$ — constant, so it is a GP.

A negative ratio means the signs alternate: positive, negative, positive, negative, ...

Next three terms: $-135 \times (-3) = 405$, $405 \times (-3) = -1215$, $-1215 \times (-3) = 3645$.

∴ it is a GP with a = 5, r = −3; the next three terms are 405, −1215, 3645.

Discuss 讨论. (1) Is 1, 4, 9, 16, ... a GP? (No — the ratios 4, 2.25, 1.78, ... are not constant; it is the square numbers.) (2) Is 1, 1, 1, 1, ... a GP? (Yes — every ratio is 1, so r = 1; a constant sequence is geometric.) (3) In Example 3, "multiply by −3" — why do the signs alternate? (A negative multiplier flips the sign at every step.)

Watch out 易错点. Always divide **term ÷ previous term**, in that order. Dividing the other way (6 ÷ 12) gives the reciprocal of r, and the rest of the working goes wrong.

3. Finding the common ratio — 求公比 r

Method 方法. (a) Consecutive terms: $r = T_{n+1} \div T_n$ — divide any term by the one before it. **(b) Two non-consecutive terms T_m and T_n:** $\frac{T_m}{T_n} = r^{m-n}$ — take the (m − n)th root. If m − n is **even**, remember r = ± root.

Example 4 — from consecutive terms

Find the common ratio of 2, 10, 50, 250, ...

$r = \frac{10}{2} = 5$ (check: $\frac{50}{10} = 5$, $\frac{250}{50} = 5$ √)

∴ r = 5.

Example 5 — from two non-consecutive terms (odd power)

A GP has $T_2 = 12$ and $T_5 = 324$. Find the common ratio and the first term.

Divide: $\frac{T_5}{T_2} = r^{5-2} = r^3 \implies r^3 = \frac{324}{12} = 27 \implies r = 3$ (since $3^3 = 27$)

First term: $a = T_2 \div r = 12 \div 3 = 4$

Check: 4, 12, 36, 108, 324 √

∴ r = 3 and a = 4.

Example 6 — from two non-consecutive terms (even power → two ratios)

A GP has $T_2 = \frac{1}{3}$ and $T_6 = 27$. Find the common ratio.

$\frac{T_6}{T_2} = r^{6-2} = r^4 \implies r^4 = 27 \div \frac{1}{3} = 81 \implies r = \pm 3$ (an even power gives two values)

If r = 3: $\frac{1}{9}, \frac{1}{3}$, 1, 3, 9, 27 √ If r = −3: $-\frac{1}{9}, \frac{1}{3}$, −1, 3, −9, 27 √

∴ r = 3 or r = −3 — both sequences reach $T_6 = 27$.

Discuss 讨论. (1) Why are there two answers in Example 6 but one in Example 5? (The power was even: $r^4 = 81$ has two real solutions, r = ±3.) (2) In Example 6, which ratio gives an alternating GP? (r = −3.) (3) What would $\frac{T_2}{T_6}$ have given? ($r^{-4} = \frac{1}{81}$, i.e. the same ±3 — so always divide later term ÷ earlier term to keep the index positive.)

Watch out 易错点. The index of r is the **difference of the term numbers** (5 − 2 = 3, 6 − 2 = 4), not the term numbers themselves. And when the power is even, **do not drop the negative ratio**.

4. Finding a term — 求第 n 项（通项）

Formula 公式 (Reference sheet). $T_n = ar^{n-1}$ — first term a, ratio r; the index is n − 1 because the first term has had **no** multiplication yet.

Example 7 — find a given term

For the GP 4, 12, 36, ..., find the 8th term.

$a = 4$, $r = \frac{12}{4} = 3$, so $T_n = 4 \times 3^{n-1}$.

$T_8 = 4 \times 3^{8-1} = 4 \times 3^7 = 4 \times 2187 = 8748$

∴ the 8th term is 8748.

Example 8 — first find a and r, then a distant term

A GP has $T_3 = 20$ and $T_6 = 160$. Find the 10th term.

$\frac{T_6}{T_3} = r^3 = \frac{160}{20} = 8 \implies r = 2$

$a = T_3 \div r^2 = 20 \div 4 = 5$ (the sequence so far: 5, 10, 20, 40, 80, 160, ... √)

$T_{10} = 5 \times 2^{10-1} = 5 \times 512 = 2560$

∴ the 10th term is 2560.

Example 9 — which term equals a given number?

Which term of the GP 3, 6, 12, 24, ... is equal to 3072?

$a = 3$, $r = 2$, so solve $3 \times 2^{n-1} = 3072$

$2^{n-1} = 1024 = 2^{10} \implies n - 1 = 10 \implies n = 11$

Check: $T_{11} = 3 \times 2^{10} = 3 \times 1024 = 3072$ √ (log method: $n - 1 = \log 1024 \div \log 2 = 10$.)

∴ the 11th term is 3072.

Discuss 讨论. (1) Why is the index n − 1 and not n? (The first term is not multiplied by anything.) (2) In Example 9, why must we write 1024 as a power of 2? (Equal bases: if $2^k = 2^m$ then k = m — this works exactly here because 1024 is a power of 2; otherwise use logarithms.) (3) How do you check an answer like 8748 quickly? (Divide by the previous term and see you get r.)

Watch out 易错点. Two classic slips: using n instead of n − 1 (the first term would come out multiplied — wrong), and losing track of the count when a question asks "which term ...?" — always finish by substituting back.

5. Sum of terms — 前 n 项和

Formulas 公式 (Reference sheet). $S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}$ $(r \neq 1)$ · If $r = 1$ the sequence is constant, so $S_n = na$. · Limiting sum: $S_\infty = \frac{a}{1-r}$ $(|r| < 1)$.

Example 10 — a doubling GP

Find the sum of the first 6 terms of $2 + 6 + 18 + \ldots$

$a = 2,\ r = 3,\ n = 6$

$S_6 = \frac{2(3^6-1)}{3-1} = \frac{2(729-1)}{2} = 728$

Check by adding: $2 + 6 + 18 + 54 + 162 + 486 = 728$ ✓

$\therefore S_6 = 728.$

Example 11 — a halving GP

Find the sum of the first 8 terms of $100 + 50 + 25 + \ldots$

$a = 100,\ r = ½,\ n = 8$ (using $S_n = a(1 - r^n) \div (1 - r)$ keeps everything positive)

$S_8 = \frac{100(1-(½)^8)}{1-½} = \frac{100(1-\frac{1}{256})}{½} = 200 \times \frac{255}{256} = \frac{6375}{32} = 199\frac{7}{32} \approx 199.22$

Check by adding: $100 + 50 + 25 + 12.5 + 6.25 + 3.125 + 1.5625 + 0.78125 = 199.218\,75$ ✓

$\therefore S_8 = 199\frac{7}{32} \approx 199.22.$

Example 12 — an alternating GP (negative ratio)

Find the sum of the first 7 terms of $1 - 2 + 4 - 8 + \ldots$

$a = 1,\ r = -2,\ n = 7$

$S_7 = \frac{1(1-(-2)^7)}{1-(-2)} = \frac{1-(-128)}{3} = \frac{129}{3} = 43$

Check by adding: $1 - 2 + 4 - 8 + 16 - 32 + 64 = 43$ ✓

$\therefore S_7 = 43.$

Extension — the infinite (limiting) sum S_∞

Find the limiting sum of $8 + 4 + 2 + 1 + \ldots$

$|r| = ½ < 1$, so the limiting sum exists: $S_\infty = \frac{8}{1-½} = 16$

The partial sums 8, 12, 14, 15, 15.5, … get closer and closer to 16 without ever passing it.

$\therefore S_\infty = 16.$

Discuss 讨论. (1) Why must $r \neq 1$ in the formula? (Both denominators would be zero; instead $S_n = na$, e.g. $1 + 1 + 1 + \ldots$) (2) Why does S_∞ exist only for $|r| < 1$? (Only then do the terms shrink towards 0; e.g. $1 + 2 + 4 + \ldots$ grows without bound — divergent.) (3) Class problem 课堂练习: a rumour is told to 2 new people in week 1, then 4, then 8, … How many people hear it in the first 5 weeks? ($a = 2$, $r = 2$: $S_5 = 2(2^5 - 1) \div (2 - 1) = 62$ people.)

Watch out 易错点. Use the $a(1 - r^n) \div (1 - r)$ form when $r < 1$ and the $a(r^n - 1) \div (r - 1)$ form when $r > 1$ — both give the same answer, but the arithmetic stays positive. And never use S_∞ unless $|r| < 1$.

6. Wrap-up — mixed discussion questions 综合练习（附答案）

Run these as quick whole-class questions: display the question, take 30–60 seconds of pair talk, then reveal the working. Answers and talking points for the teacher follow each question.

Question 问题	Answer and teaching point 答案与要点
1. Is 0.5, 1.5, 4.5, 13.5, ... a GP? If so, state r.	Yes: $\frac{1.5}{0.5} = \frac{4.5}{1.5} = \frac{13.5}{4.5} = 3$, so $\mathbf{r = 3}$. (Point: check at least two ratios before saying yes.)
2. Find the common ratio of 81, −27, 9, −3, ...	$\mathbf{r = -\frac{1}{3}}$ ($-27 \div 81 = -\frac{1}{3}$). (Alternating GP — signs flip each term.)
3. A GP has $T_4 = 40$ and $T_7 = 320$. Find r, a and the 9th term.	$r^3 = \frac{320}{40} = 8 \Longrightarrow \mathbf{r = 2}$; $a = 40 \div 2^3 = \mathbf{5}$; $T_9 = 5 \times 2^8 = \mathbf{1280}$. (Sequence 5, 10, 20, 40, 80, 160, 320, ...)
4. Which term of 5, 15, 45, ... is equal to 3645?	$5 \times 3^{n-1} = 3645 \Longrightarrow 3^{n-1} = 729 = 3^6 \Longrightarrow n - 1 = 6 \Longrightarrow \mathbf{n = 7}$ — the 7th term. (Check: $5 \times 3^6 = 3645$ √)
5. Find the sum of the first 6 terms of ½ + 1 + 2 + 4 + ...	$a = ½$, $r = 2$: $S_6 = \frac{½(2^6 - 1)}{2 - 1} = \mathbf{31.5}$. (Check: 0.5 + 1 + 2 + 4 + 8 + 16 = 31.5 √)
6. Find the limiting sum of 27 + 9 + 3 + ...	$\lvert r\rvert = \frac{1}{3} < 1$, so $S_\infty = 27 \div (1 - \frac{1}{3}) = 27 \times \frac{3}{2} = \mathbf{40.5} = \mathbf{40½}$. (Terms shrink to 0, so the sum settles.)
7. For 1 + (−1) + 1 + (−1) + ... : does S_∞ exist? What is S_{50}?	No — $\lvert r\rvert = 1$ is **not** less than 1, so there is no limiting sum. But the 50 terms form 25 pairs (1 − 1), so $\mathbf{S_{50} = 0}$. (Great contrast question: finite sums can exist even when S_∞ does not.)

Formula summary — 公式小结. nth term: $T_n = ar^{n-1}$ · ratio from terms m and n: $r^{m-n} = T_m \div T_n$ · sum: $S_n = a(1 - r^n) \div (1 - r) = a(r^n - 1) \div (r - 1)$ · limiting sum: $S_\infty = a \div (1 - r)$, $|r| < 1$ · three-term test: $T_2^2 = T_1 T_3$.